

QUESTION BANK

1. Solve the equation $\frac{dy}{dx} = x^2 + y$, given $y(0) = 1$, $h=0.05$ by using Modified Euler's method. Calculate y at $x=0.1$ (Up to four decimal places)

2. Use Runge-Kutta method of fourth order to solve the equation

$$\frac{dy}{dx} = x^2 + y^2, \text{ subject to } y(1) = 1.5. \text{ Calculate } y \text{ at } x=1.2 \text{ taking } h=0.1$$

3. Use Runge-Kutta method of fourth order to find $y(0.1)$ and $y(0.2)$

$$\frac{dy}{dx} = x - y, \quad y(0) = 0, \text{ taking } h = 0.1.$$

4. Fit a parabola of the form $y = ax^2 + bx + c$ to the following data:

X	1	2	3	4	5	6
Y	4	9	16	25	36	49

Use method of least squares.

5. Solve $\frac{dy}{dx} = x + y^2$, $y(0) = 1$ by using Euler's method. Find y at

$$x = 0:0.1:0.9, \quad h = 0.1$$

FOR PRACTICE Evaluate $\int_0^1 \frac{1}{1+x^2} dx$ by dividing the range into 6 equal parts using Simpson's $1/3^{\text{rd}}$ rule.

6. Evaluate $\int_0^5 \frac{1}{1+x} dx$ by using Trapezoidal rule. (Take $h=0.5$)

7. Evaluate $\int_0^{0.8} (\log_e(x+1) + \sin \sin 2x) dx$ where x is in radian. Use Simpson's $\frac{1}{3}^{\text{rd}}$ rule,
taking $h = 0.1$

8. The set of points (x_i, y_i) , $i = 0, 1, 2, 3$ are $(0, 1), (1, 1), (2, 7), (3, 25)$, then the value of $\Delta^2 y_0$ is _____

9. If the Lagrange's polynomial passes through

X	0	2
y	-3	1

Then $\int_0^2 y \, dx$ is equal to -----

10. Using Newton's Forward interpolation formula, find y at $x = 2.2$

x	0	1	2	3	4	5
y	1	3	7	13	21	31

11. Find the value of y at $x = 2$, by using Lagrange's interpolation formula.

X	1	3	5	6
Y	2	10	26	37

12. Compute $\cosh \cosh (4.15)$, using appropriate interpolation formula to find the value of y at $x = 4.25$

x	4.0	4.1	4.2	4.3	4.4	4.5
Cosh x	27.3081	30.1784	33.3507	36.8567	40.7316	45.0141

13. Following table shows the enthalpy at different pressures

Pressure(Bar)	1.9	2.2	2.4	2.6
Enthalpy (KJ/Kg.k)	497.9	517.6	529.6	540.9

Find out the enthalpy at pressure 2.1 bar by Lagrange's interpolation method.--

14. Obtain regression lines for the following data

x	25	30	32	35	37	40	42	45
y	8	10	15	17	20	23	24	25

15. If the probability that an individual suffers a bad reaction from a certain injection is 0.001, determine the probability that out of 2000 individuals (i) exactly 3 (ii) more than two individuals (iii) more than one individual will suffer a bad reaction.
16. In a certain examination test, 2000 students appeared in a subject of statistics.
17. Average marks obtained were 50% with standard deviation 5%. How many students do you expect to obtain more than 60% of marks, Supposing that marks are distributed normally? [Given $z=2$, Area=0.4772]
18. The first four moments about the working mean 4 of a distribution are 1, 4, 10 and 45. Calculate the first four moments about the mean. Also evaluate β_1 , β_2

19. The probability that a man aged 60 will live to be 70 is 0.65. What is the probability that out of 10 men (i) at most two (ii) at least seven will live to be 70?
20. Find the mean , mode, median for the following data.

x	1	3	2	5	6
y	3	7	5	11	13

21. Prepare Forward difference table for the following data

X	0	1	2	3	4	5
Y	3	4	11	30	67	128

Find the value of second order forward difference of y_1

22. Prepare a Backward difference table for the following data

X	1	3	5	7	9	11
Y	2	10	26	50	82	122

23. Show that

i.
$$\frac{\Delta}{\nabla} - \frac{\nabla}{\Delta} = E - E^{-1}$$

ii.
$$(E^{\frac{1}{2}} + E^{-\frac{1}{2}})(1 + \Delta)^{\frac{1}{2}} = 2 + \Delta$$

24. If $f(x) = x^2 + 5x + 2$, $h = 1$ then find

$$\Delta^2 f(x), \quad \nabla(f(x)), \quad E^{12}(f(x)) \text{ \& } \delta(f(x))$$

EXERCISE 58.1

1. Find the linear least square polynomials based on data

x	-2	-1	0	1
y	6	3	2	2

is given. Find the least square straight line approximation to the data. **Ans.** $10y = 26 - 13x$

2. The following table shows the number of salesmen working for a certain concern :

Year	1998	1999	2000	2001	2002	2003
Number	28	38	46	40	56	60

Use the method of least squares to fit a straight line trend. **Ans.** $y = 5.9428x - 11843.9047$

1. Calculate the coefficient of correlation from the data given below :

x	4	6	8	10	12
y	2	3	4	6	10

Ans. 0.95

3. Find the coefficient of correlation of the following data taking new origin of x at 70 and for y at 67:

x	67	68	64	68	72	70	69	70
y	65	66	67	67	68	69	71	73

(A.M.I.E., Winter 2002) **Ans.** 0.472

1. Find the values of a , b , c so that $y = a + bx + cx^2$ is the best fit to the data :

x	0	1	2	3	4
y	1	0	3	10	21

Ans. $a = 1, b = -3, c = 2$

2. Fit a second degree parabola to the following data by Least squares method :

x	0	1	2	3	4
y	1	1.8	1.3	2.5	6.3

Ans. $y = 1.42 - 1.07x + 0.55x^2$

3. Fit a second degree parabola to the following data taking y as dependent variable :

x	1	2	3	4	5	6	7	8	9
y	2	6	7	8	10	11	11	10	9

Ans. $y = -1 + 3.55x - 0.27x^2$

4. Use the least-square method to obtain a parabola that approximates the data :

x	1.0	1.2	1.4	1.6	1.8	2
y	2.345	2.419	2.592	2.863	3.233	3.702

Ans. $y = 3.124 - 2.477x + 1.458x^2$

Two judges in a beauty contest rank the ten competitors in the following order :

6	4	3	1	2	7	9	8	10	5
4	1	6	7	5	8	10	9	3	2

Do the two judges appear to agree in their standard ?

1. Find the regression line of y on x for the data :

x	1	4	2	3	5
y	3	1	2	5	4

Ans. $y = 2.7 + 0.1x$

2. Compute the regression line of x on y for the following data :

x	2	4	6	8	10
y	12	10	8	6	4

Ans. $x = 1 + 3y$

3. Compute the regression line of y on x for the following data :

x	1	2	3	4	5	6
y	2	2	2	2	2	2

Ans. $y = 6 - x$

4. Find the regression lines of from the given data :

x	1	2	3	4	5	6	7	8	9	10
y	10	12	16	28	25	36	41	49	40	50

Ans. $x = 0.2y - 0.64$, $y = 4.69x + 4.9$

5. Find the equations to the lines of regression and the coefficient of correlation for the following data:

x	2	4	5	6	8	11
y	18	12	10	8	7	5

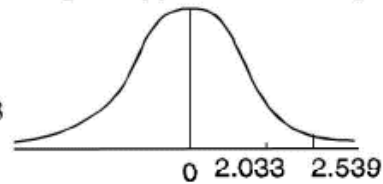
Ans. $y - 10 = -1.34(x - 6)$, $x - 6 = -0.632(y - 10)$, $r = -0.92$

Example 7. A manufacturer intends that his electric bulbs have a life of 1000 hours. He tests a sample of 20 bulbs, drawn at random from a batch and discovers that the mean life of the sample bulbs is 990 hours with a S.D of 22 hours. Does this signify that the batch is not up to the standard?

[Given: The table value of t at 1% level is significance with 19 degrees of freedom is 2.539]

Solution. $\bar{x} = 990$, $\sigma = 22$, $x = 1000$

$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}} = \frac{990 - 1000}{\frac{22}{\sqrt{20}}} = -\frac{10\sqrt{20}}{22} = -\frac{22.36}{11} = -2.033$$



Since the calculated value of t (2.032) is less than the value of t (2.539) from the table. Hence, it is not correct to say that this batch is not upto this standard.

Ans.

- Find the probability that at most 5 defective fuses will be found in a box of 200 fuses if experience shows that 2 per cent of such fuses are defective.
Ans. 0.785
- The number of accidents during a year in a factory has the Poisson distribution with mean 1.5. The accidents during different years are assumed independent. Find the probability that only 2 accidents take place during 2 years time.
Ans. 0.224
- A manufacturer of cotter pins knows that 5% of his product is defective. If he sells cotter pins in boxes of 100 and guarantee that not more than 10 pins will be defective, what is the approximate probability that a box will fail to meet the guaranteed quality. [$e^{-5} = 0.006738$]
Ans. 0.0136875
- Suppose the number of telephone calls on an operator received from 9.00 to 9.05 follow a Poisson distribution with mean 3. Find the probability that
 - the operator will receive no calls in that time interval tomorrow,
 - in the next three days the operator will receive a total of 1 call in that time interval.**Ans.** (i) e^{-3} (ii) $3 \times (e^{-3})^2 (e^{-3} \cdot 3)$
[$e^{-3} = 0.04978$]
- On the basis of past record it has been found that there is 70% chance of power cut in a city on any particular day. What is the probability that from the first to the 10th date of the month, there are 5 or more days without power cut.
(A.M.I.E.T.E., Summer 2001)

$$\text{Ans. } \left[\frac{3^5}{5!} + \frac{3^6}{6!} + \frac{3^7}{7!} + \frac{3^8}{8!} + \frac{3^9}{9!} + \frac{3^{10}}{10!} \right] e^{-3}$$

Example 14. Fit a Binomial distribution for the following data and compare the theoretical frequencies with actual ones :

x	0	1	2	3	4	5
y	2	14	20	34	22	8

(R.G.P.V., Bhopal, II Semester, June 2006)

Solution.

x	$y = f$	fx	$P = {}^5C_x p^x q^{5-x}$	Theoretical Frequency
0	2	0	${}^5C_0 (0.568)^0 (0.432)^5 = 0.015$	$100 \times 0.015 = 1.5$
1	14	14	${}^5C_1 (0.568)^1 (0.432)^4 = 0.099$	$100 \times 0.099 = 9.9$
2	20	40	${}^5C_2 (0.568)^2 (0.432)^3 = 0.260$	$100 \times 0.260 = 26.0$
3	34	102	${}^5C_3 (0.568)^3 (0.432)^2 = 0.342$	$100 \times 0.342 = 34.2$
4	22	88	${}^5C_4 (0.568)^4 (0.432)^1 = 0.225$	$100 \times 0.225 = 22.5$
5	8	40	${}^5C_5 (0.568)^5 (0.432)^0 = 0.0591$	$100 \times 0.0591 = 5.91$
	100	284		

$$\Sigma f = 100,$$

$$\Sigma fx = 284$$

$$\text{Mean} = \frac{\Sigma fx}{\Sigma f} = \frac{284}{100} = 2.84$$

$$\text{Mean} = np = 2.84$$

$$5p = 2.84 \Rightarrow p = \frac{2.84}{5} = 0.568$$

$$q = 1 - p = 1 - 0.568 = 0.432$$

$$\text{Binomial Distribution} = 100 (0.432 + 0.568)^5$$

Ans.

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1. In a regiment of 1000, the mean height of the soldiers is 68.12 units and the standard deviation is 3.374 units. Assuming a normal distribution, how many soldiers could be expected to be more than 72 units? It is given that
 $P(z = 1.00) = 0.3413$, $P(z = 1.15) = 0.3749$ and
 $P(z = 1.25) = 0.3944$, where z is the standard normal variable. **Ans.** 125
 2. The lifetime of radio tubes manufactured in a factory is known to have an average value of 10 years. Find the probability that the lifetime of a tube taken randomly (i) exceeds 15 years, (ii) is less than 5 years, assuming that the exponential probability law is followed. **Ans.** (i) 0.2231, (ii) 0.3935.
 3. The breaking strength X of a cotton fabric is normally distributed with $E(x) = 16$ and $\sigma(x) = 1$. The fabric is said to be good if $X \geq 14$. What is the probability that a fabric chosen at random is good. Given that $\phi(2) = 0.9772$ **Ans.** 0.9772
 4. A manufacturer knows from experience that the resistance of resistors he produces is normal with mean $\mu = 140 \Omega$ and standard deviation $\sigma = 5\Omega$. Find the percentage of the resistors that will have resistance between 138Ω and 142Ω . (given $\phi(0.4) = 0.6554$, where z is the standard normal variate). **Ans.** 31.08%
 5. A manufacturing company packs pencils in fancy plastic boxes. The length of the pencils is normally distributed with $\mu = 6''$ and $\sigma = 0.2''$. The internal length of the boxes is $6.4''$. What is the probability that the box would be too small for the pencils (Given that a value of the standardized normal distribution function is $\phi(2) = 0.9772$). **Ans.** 0.0228.

Solution of algebraic and transcendental equations

Find a positive root of the equation $e^x \cos x - 1.2 = 0$, at the end of third iteration using Newton- Raphson Method.

[Initial value $x_0 = 1.2$]

Find the root of $x^3 + 2x - 5 = 0$ by using Newton-Raphson method.

(Perform three iterations)

Find the root of $x^2 - 25 = 0$ by using Newton-Raphson method.
(Perform three iterations)

Find the root of $x^3 - 16 = 0$ by using Newton-Raphson method.
(Perform three iterations)

Find real root of the equation $e^x - 5x = 0$ by using method of False position. (Perform three iterations)

Find real root of $8x^3 - 6x - 1 = 0$, using Successive approximation method. (Take initial approximation as 0.95) (Perform three iterations)

Find real root of $x^3 + 2x + 1 = 0$, using Successive approximation method. (Perform three iterations)

Find real root of $x^2 - x - 2 = 0$, using Successive approximation method. (Perform three iterations)

Find the root of $x - x \cos \cos x = 0$ by using Newton-Raphson method. (Perform three iterations)

Find real root of the equation $x^3 - 5x + 3 = 0$ by using method of False position. (Perform three iterations)

Find real root of the equation $x^3 - 2x - 5 = 0$ by using method of False position. (Perform three iterations)

Newton Interpolation

Find the value of y at $x = 4.5$ and $\frac{dy}{dx}$ at $x=1.5$ for the following data, using appropriate difference formulae.

x	1	2	3	4	5
y	3.47	6.92	11.25	16.75	22.949

Evaluate $\frac{dy}{dx}$ at $x=1.5$ for the following data, using appropriate difference formulae.

x	1	2	3	4
y	3	6	11	18

Ans:- $y=x^2+2$

Evaluate $\frac{dy}{dx}$ at $x=4.5$ for the following data, using appropriate difference formulae.

x	2	3	4	5
y	9	19	33	51

Ans:- $y=2x^2+1$

The Standard Deviation and Arithmetic Mean of three distribution X,Y,Z are as follows:

	Arithmetic Mean	Standard deviation
X	18.0	5.4
Y	22.5	4.5
Z	24.0	6.0

The more stable distribution is

ANS- Y

The Standard Deviation and Arithmetic Mean of three distribution X,Y,Z are as follows:

	Arithmetic Mean	Standard deviation
X	15	2.4
Y	24	4.5
Z	25	2.2

The more stable distribution is

Ans-Z

A die is tossed once . Random variable x denote the digits that appeared. The value of expectation of x is

- a. $7/2$
- b. $3/2$
- c. 1
- d. None of these

There are three packets containing 100 rupees, 400 rupees and 700 rupees respectively. A player selects an envelope and keeps with him, what he gets. Then the expected gain of the player is

- a. 100
- b. 400
- c. 700
- d. 3

A probability distribution table is as follows

X	1	2	3
P(x)	$1/6$	$2/3$	$1/6$

The value of $E(x^2)$ is-----

- a. $13/3$
- b. 4
- c. $16/3$
- d. 5

Given density function

$$f(x) = ke^{-ax}, \quad x \geq 0, a > 0$$

$$= 0, \quad \text{otherwise}$$

The value of k is

- a. a
- b. 1
- c. 0.5
- d. None of these

A continuous random variable x has the pdf defined as

$$p(x) = ke^{-|x|}, -\infty < x < \infty$$

The value of k is

- a. $\frac{1}{2}$
- b. $\frac{1}{3}$
- c. 1
- d. $\frac{3}{4}$

Find the value of k if the probability distribution function is

X	1	2	3
P(x)	k	2k	3k

- a. $\frac{1}{3}$
- b. $\frac{1}{8}$
- c. $\frac{3}{8}$
- d. None of these

If $E(x)=3$ and $E(x^2)=12$ then the variance of x is

- a. 4
- b. 11
- c. 9
- d. 1

If 4 children are there in a family with $p=1/2$ and $q=1/2$, the probability of at least one boy is

- a. 15/16
- b. 11/16
- c. 1/16
- d. None of these

If the probability density function $f(x)$ of a continuous random variable x is defined as

$$f(x) = \begin{cases} \frac{3}{2}x^2, & -1 \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases} \text{ then } P(0 \leq x \leq 1) \text{ is}$$

- a. $\frac{1}{2}$
- b. 0
- c. $\frac{3}{7}$
- d.